

Algebra 1
Unit 5
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

Choose numbers from the positive digits 1 – 9 to fill in the boxes below for questions 1–4. *You may only use each digit one time.*

$$y = \frac{\boxed{3}}{\boxed{4}} \cdot x + \boxed{5}$$

1. List the variables in your equation.

x, y

2. List the parameters in your equation.

Answers will vary.

$\frac{3}{4}$

3. Choose new values for the parameters so that the value of y will be as **large** as possible if $x = 1$. *You will still use the digits 1 – 9 only one time, but you may re-use numbers from the previous equation.* How do you know your parameters make y as large as possible?

$$y = \frac{8}{1}x + 9$$

9 is the largest starting point and $\frac{8}{1}$ is largest slope without repeating digits.

4. Choose new values for the parameters so that the value of y will be as **small** as possible if $x = 1$. *You will still use the digits 1 – 9 only one time, but you may re-use numbers from the previous equation.* How do you know your parameters make y as small as possible?

$$y = \frac{2}{9}x + 1$$

1 is the smallest starting point and $\frac{2}{9}$ is smallest slope without repeating digits.

5. Xander looked at the equation $3y + 4 = 9x - 11$ and said, "The variables are x and y . The parameters are 3, 4, 9, and -11 ." Z'Kera said, "I agree about the variables, but the parameters are 3 and 5." Who is correct, Xander, Z'Kera, or both?

Neither. When the equation is put into slope-intercept form it is $y = 3x - 5$. The slope/parameter is 3.

Yvonne earns a fixed amount of money per day as a substitute teacher and earns an hourly rate for each hour that she works as an after-school tutor each week. Yvonne's pay can be modeled by the function $f(x) = 15x + 100$.

6. Which value(s) in the equation are variables?
 $f(x)$ and x
7. Which value(s) in the equation are parameters?
 15
8. What does the value 15 mean in terms of Yvonne's pay?
Yvonne makes \$15 per hour.
9. What does the value 100 mean in terms of Yvonne's pay?
It is the fixed pay rate for substitute teaching.
10. Complete the Venn Diagram to compare and contrast variables and parameters.

